Preface

Who is This Book For?

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Why Write Yet Another Book About Linear Algebra?

The direct answer to this question is simple: I really think most resources are doing a *bad job* of explaining fundamental linear algebra to new students [1]. Not because they lack material or the complete picture - but because the common approach to teaching fundamental mathematical topics just doesn't work well for most people [2].

Introduction courses to fundamental mathematical topics usually look like this: here'e are some basic definitions, let's derive some specific results from them - now go apply these in your field.

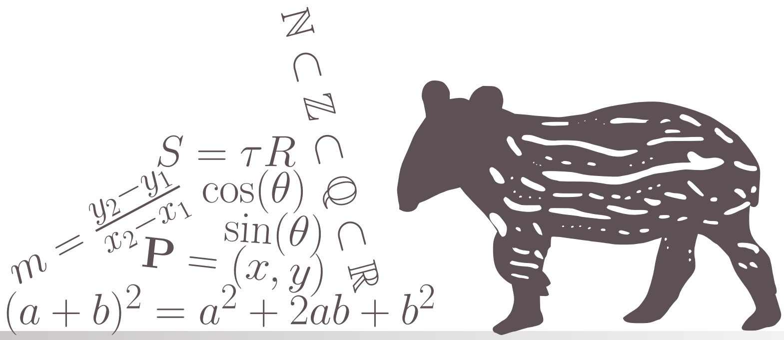
[1] and honestly, this isn't restricted to linear algebra but applies to all fundamental mathematical topics - but it's a classic case

[2] the main exception is students of pure mathematics and people who can easily deal with extremely abstract concepts - a very specific group of people at the end of the day

CHAPTER

1

THE BASICS OF LINEAR ALGEBRA



1.1 VECTORS AND VECTOR SPACES

1.1.1 Motivation

Motivating Problem 1.1 Gravity Force

Consider a simple simulation of classical ("Newtionian") gravity, say of a planet with mass m orbiting a star with mass M (where $M \gg m$). At any given moment, the force acting on the planet due to the star's gravity has magnitude

$$F = G\frac{Mm}{r^2},$$

where r is the distance between the planet and the star at that moment, and G is a universal constant (specifically, $G = 6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$).

However, since we live in 3-dimensional space[a], we also want this force to be pointing in the direction from the planet to the star. To do this, we need to understant how to describe orientations in 3-dimensional space, and be able to pack together both orientation and magnitude in the correct way.

[a] allegedly.

Motivating Problem 1.2 Bouncing/Reflecting Off a Wall

Imagine we are writing a 3D computer game in which some object is bouncing off a wall. Alternatively, consider 3D graphics in general, where one simulates rays of light reflecting off of smooth surfaces. In both cases the same problem needs to be solved: given a flat plane of some orientation in space, and an object hitting it from some direction - what would be the direction of the object after it bounces off the plane?

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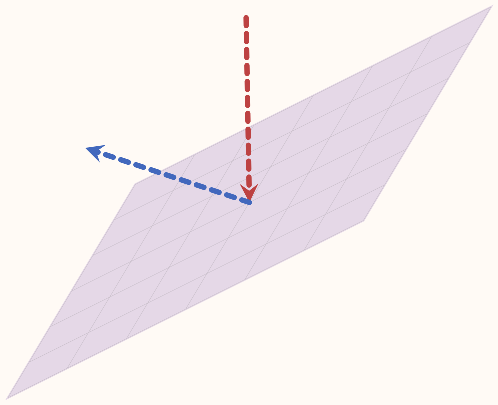

To solve this problem we will need to know how to represent quantities like position, velocity and orientation in 3D, and more generally points, lines and planes - and to know how to calculate the intersection between these different objects. By the end of this section you should be able to solve this problem by yourself, and even generalize it to any number of dimensions.

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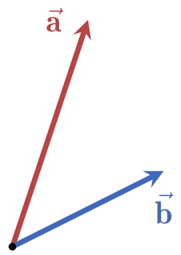
1.1.2 Geomteric Properties of Vectors

Text text text...(what are vectors, pictorial representation, notation, all starting from the origin, etc.)

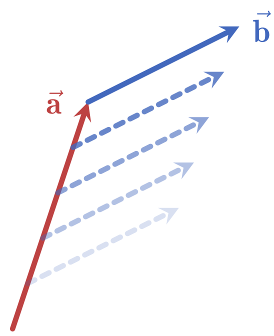
Vectors have two important properties: they can be added together and be scaled to form other vectors.

- **Adding vectors together**: any two vectors of the same dimensionality (e.g. 2-dimensional, 3-dimensional, 4-dimensional, etc.) can be added together, and the result is a third vector. Pictorially, here are two vectors $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ which we want to sum together:

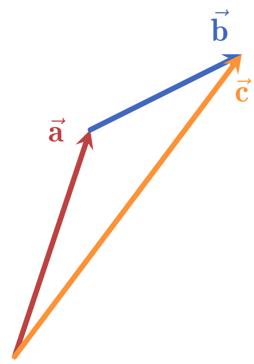


The steps to produce the sum of $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ is the following:

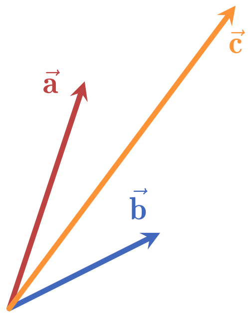
1. move $\vec{\mathbf{b}}$ such that its origin aligns with the head of $\vec{\mathbf{a}}$:



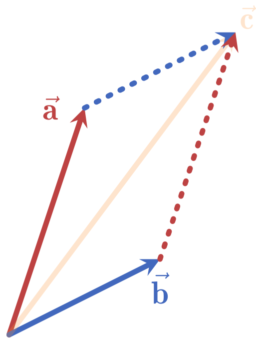
2. draw a new vector from the origin to the head of the shifted $\vec{\mathbf{b}}$:



3. the resulting vector $\vec{c}$ *is* the sum of $\vec{a}$ and $\vec{b}$. Here are all three vectors together from the same origin:

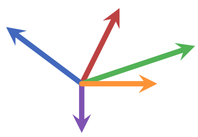


This method is called the **triangle method**, and it displays an important property of vector addition: it is commutative! This means that the order of addition doesn't matter: $\vec{a} + \vec{b} = \vec{b} + \vec{a}$. To see this visually, we can apply the triangle method twice, i.e. moving both $\vec{a}$ *and* $\vec{b}$:

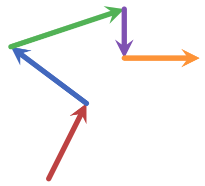


We see that no matter which vector we add onto which (i.e. what is the order of addition) - the result is the same. The shape formed by this demonstration is called a **parallelogram**, and it is incredibly important in linear algebra as a whole as we will see in future sections.

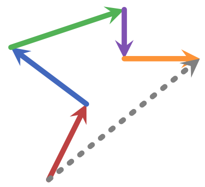
This method of summing vectors can be applied to any number of vectors. For example, consider the following 5 vectors:



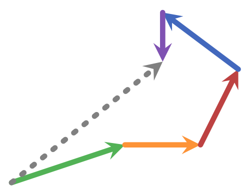
To sum them together, all we need is to start with one the vectors (say, the red one) and use the triangle method to add the other vectors step by step:



Finally, we connect the origin of the first vector with the head of the last vector (in this case the orange one):

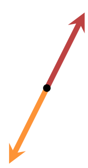


and that's it, the dashed gray vector is the sum of all five vectors. To see that the order of addition really doesn't matter we can do this again but this time starting with the green vector and changing the summation order completely:



As you can see, the resulting dashed gray vector is exactly the same as before.

Another important property of the addition of vectors is that we can always take a vector and rotate it by 180°, giving us a vector that "cancels" the original vector by summation. For example, the following red vector is rotated to yield its opposite, colored in orange:



These vectors added together using the triangle method yield a vector with 0 norm:

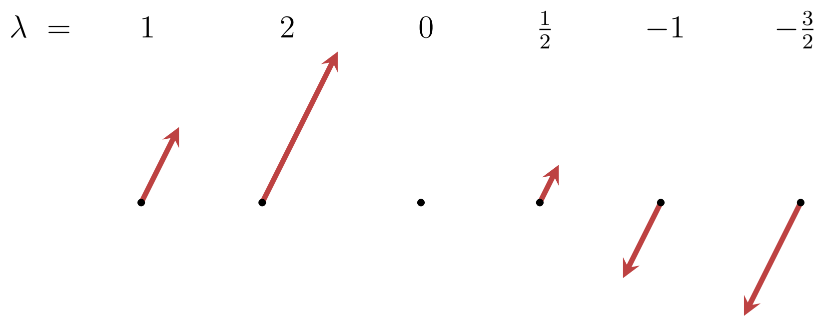
(here I drew the vectors a bit shifted from each other so that we can see both)

Such a vector is called the **zero vector**, denoted as $\vec{0}$. It has no direction, and a norm of 0. Adding $\vec{0}$ to any other vector is the same as not adding any vector at all (and it doesn't matter "from which side" we add $\vec{0}$):

$$\vec{a} + \vec{0} = \vec{0} + \vec{a} = \vec{a}. \tag{1.1}$$

We say that the zero vector is *agnostic* (also: neutral) to addition.

• **Scaling vectors**: any vector can be **scaled** by any real number, which in this context we call a **scalar**. Scaling by a number simply "stretches" the vector by the number. For example, in the following figure a red vector is scaled by a scalar λ, taking on different values:



This example shows us several things:

- no matter by which positive scalar we scale a vector, it always points in the same direction.
- If we scale a vector by 0 it becomes the zero vector.
- Finally, scaling a vector by a negative scalar rotates the vector by 180° and then scales it by the absolute value of the scalar.

This is very similar to how real numbers behave under multiplication: multiplying a real number by a negative number is, in a sense, "flipping" and scaling it relative to 0 (the origin). For example, multiplying 3 by -2 flips the the number to point to the left, and then scales it to be of length 6:

Therefore, scaling a vector is denoted just like multiplication of real numbers: $\lambda\vec{a}$, where λ is the scalar. In the above example, if the red vector is $\vec{\mathbf{a}}$, then the other five vectors can be written as $2\vec{\mathbf{a}}$, $0\vec{\mathbf{a}}$, $\frac{1}{2}\vec{\mathbf{a}}$, $-1\vec{\mathbf{a}}$ and $-\frac{3}{2}\vec{\mathbf{a}}$.

Furthermore, we can place the vector notation in different places and even hiding the scalar depending on convinience, so for example we can write $-\vec{\mathbf{a}}$ instead of $-1\vec{\mathbf{a}}$ and $-\frac{3\vec{\mathbf{a}}}{2}$ instead of $-\frac{3}{2}\vec{\mathbf{a}}$.

1.1.3 Common Coordinate Systems in 2- and 3-Dimensions

1.1.4 Geomteric Objects

1.1.5 Projections and Rejections

1.1.6 Vector Spaces

1.1.7 Solving the Motivating Problems

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